Qt 工程编译说明

Revision History

Draft Date	Revision No.	Description
2024/06/25	V1.0	1. 初始版本。

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前　言

本文档主要说明如何将已有 Qt 工程进行编译，生成可在评估板上正常运行的 ARM 端 Qt 程序镜像。

Qt 开发环境主要包括 Qt Creator 工具、X86 端 Qt 环境、ARM 端 Qt 环境、交叉编译工具链等。当用户使用 Qt Creator 工具和 X86 端 Qt 环境编写并编译 Qt 工程后，生成的 Qt 程序镜像仅可在 X86 端运行。必须使用适配评估板的 ARM 端 Qt 环境、交叉编译工具链编译 Qt 工程，生成的 Qt 程序镜像方可在 ARM 端（即评估板）上正常运行。

Linux SDK 中已提供适配评估板的 ARM 端 Qt 环境(Qt-6.3.2)和交叉编译工具链。请先参考《Linux 开发环境搭建》相关文档安装 SDK，并配置交叉编译工具链以及自行安装 Qt Creator 工具。可基于命令行或 Qt Creator 工具对已有 Qt 工程进行编译，生成 ARM 端 Qt 程序镜像。

我司提供的 Qt 工程位于产品资料“4-软件资料\Demo\qt-demos\”目录下，其中 src 目录下包含 Qt 工程源码，bin 目录下包含 ARM 端 Qt 程序镜像。请将整个 qt-demos 文件夹拷贝至 Ubuntu 工作目录下，如下图所示。

```
tronlong@tronlong-virtual-machine:~/qt-demos$ pwd
/home/tronlong/qt-demos
tronlong@tronlong-virtual-machine:~/qt-demos$ tree -L 2
.
├── image_display
│   ├── bin
│   └── src
└── led_control
    ├── bin
    └── src

6 directories, 0 files
tronlong@tronlong-virtual-machine:~/qt-demos$
```

图 1

备注：我司编译文件系统所用的 Qt 源码版本为 6.3.2，但编译生成的 Qt 库本版打印信息为 6.3.3（打印信息不准确），即实际使用 Qt 库版本应为 6.3.2，此问题不影响功能测试。

```
tronlong@tronlong-virtual-machine:~/SDK$ source environment-setup-armv8a-poky-li
nux
tronlong@tronlong-virtual-machine:~/SDK$ qmake -v
QMake version 3.1
Using Qt version 6.3.3 in /home/tronlong/SDK/sysroots/armv8a-poky-linux/usr/lib
tronlong@tronlong-virtual-machine:~/SDK$
```

图 2 Qt 库版本

```
DESCRIPTION ?= "Qt is a cross-platform application development framework for desktop, embedded and mobile."
HOMEPAGE ?= "https://www.qt.io"

QT_MODULE ?= "${BPN}"
QT_MODULE_BRANCH ?= "6.3"
QT_MODULE_BRANCH_PARAM ?= "branch=${QT_MODULE_BRANCH}"

SRC_URI = "${QT_GIT}/${QT_GIT_PROJECT}/${QT_MODULE}.git;name=${QT_MODULE};${QT_MODULE_BRANCH_PARAM};protocol=${QT_GIT_PROTOCOL}"

CVE_PRODUCT = "qt"

S = "${WORKDIR}/git"

PV = "6.3.2"

SRCREV = "${SRCREV_${QT_MODULE}}"

SRCREV_tqtc-qmlcompilerplus = "b9dc99ed1500f877642f005a44f4d618dab8ddad"
SRCREV_qt3d = "b6d9498192e589b1113d14ffb6b1c7fba2a7b468"
SRCREV_qt3d-assimp = "5a38cd0a03015ceabbd5bc6efb0730dde1ef74e5"
SRCREV_qt5compat = "6bc6fe33cebd496a529749ddd7a539010aedd838"
SRCREV_qtbase = "fe0dd828ab031eb1fb6bcf081b463833086c5383"
SRCREV_qtapplicationmanager = "9dbcab30dd87b4710f985ab09f1d5a8431dfe312"
SRCREV_qtcharts = "5e0013b27e7846c1e03a173e5929583803618288"
SRCREV_qtcoap = "28cdb946d3c25a4aebc6e629c071b501a902b4dd"
SRCREV_qtconnectivity = "d369d77c8ec3cae31a7b08541958507132dd6036"
SRCREV_qtdatavis3d = "113e1124b7d4c9d8b82b3d7b2c180bd98e122a29"
```

图 3 Qt 源码版本

1 基于命令行编译 Qt 工程

进入 image_display 工程源码目录，配置交叉编译工具链环境变量，执行 qmake 命令生成 Makefile 文件，再执行 make 命令编译生成可在评估板上正常运行的 ARM 端 Qt 程序镜像，如下图所示。

Host# source /home/tronlong/SDK/environment-setup-armv8a-poky-linux

Host# qmake

Host# make

```
tronlong@tronlong-virtual-machine:~/qt-demo/image_display/src$ pwd
/home/tronlong/qt-demo/image_display/src
tronlong@tronlong-virtual-machine:~/qt-demo/image_display/src$ source /home/tron
long/SDK/environment-setup-armv8a-poky-linux
tronlong@tronlong-virtual-machine:~/qt-demo/image_display/src$ qmake
Info: creating stash file /home/tronlong/qt-demo/image_display/src/.qmake.stash
tronlong@tronlong-virtual-machine:~/qt-demo/image_display/src$ make
aarch64-poky-linux-g++  -march=armv8-a+crc+crypto -fstack-protector-strong  -O2
-D_FORTIFY_SOURCE=2 -Wformat -Wformat-security -Werror=format-security --sysroot
=/home/tronlong/SDK/sysroots/armv8a-poky-linux -c -pipe  -O2 -pipe -g -feliminat
e-unused-debug-types  -O2 -Wall -Wextra -D_REENTRANT -fPIC -DQT_DEPRECATED_WARNI
NGS -DQT_NO_DEBUG -DQT_WIDGETS_LIB -DQT_GUI_LIB -DQT_CORE_LIB -I. -I../../../SDK
/sysroots/armv8a-poky-linux/usr/include/QtWidgets -I../../../SDK/sysroots/armv8a
-poky-linux/usr/include/QtGui -I../../../SDK/sysroots/armv8a-poky-linux/usr/incl
ude/QtCore -I. -I../../../SDK/sysroots/armv8a-poky-linux/usr/lib/mkspecs/linux-o
e-g++ -o main.o main.cpp
aarch64-poky-linux-g++  -march=armv8-a+crc+crypto -fstack-protector-strong  -O2
-D_FORTIFY_SOURCE=2 -Wformat -Wformat-security -Werror=format-security --sysroot
=/home/tronlong/SDK/sysroots/armv8a-poky-linux -c -pipe  -O2 -pipe -g -feliminat
e-unused-debug-types  -O2 -Wall -Wextra -D_REENTRANT -fPIC -DQT_DEPRECATED_WARNI
NGS -DQT_NO_DEBUG -DQT_WIDGETS_LIB -DQT_GUI_LIB -DQT_CORE_LIB -I. -I../../../SDK
/sysroots/armv8a-poky-linux/usr/include/QtWidgets -I../../../SDK/sysroots/armv8a
-poky-linux/usr/include/QtGui -I../../../SDK/sysroots/armv8a-poky-linux/usr/incl
ude/QtCore -I. -I../../../SDK/sysroots/armv8a-poky-linux/usr/lib/mkspecs/linux-o
e-g++ -o mainwindow.o mainwindow.cpp
aarch64-poky-linux-g++  -march=armv8-a+crc+crypto -fstack-protector-strong  -O2
-D_FORTIFY_SOURCE=2 -Wformat -Wformat-security -Werror=format-security --sysroot
=/home/tronlong/SDK/sysroots/armv8a-poky-linux -pipe  -O2 -pipe -g -feliminate-u
nused-debug-types  -O2 -Wall -Wextra -dM -E -o moc_predefs.h ../../../SDK/sysroo
ts/armv8a-poky-linux/usr/lib/mkspecs/features/data/dummy.cpp
```

图 4

```
tronlong@tronlong-virtual-machine:~/qt-demo/image_display/src$ ls
image_display      main.o          mainwindow.o        moc_mainwindow.o
image_display.pro  mainwindow.cpp  Makefile            moc_predefs.h
main.cpp           mainwindow.h    moc_mainwindow.cpp
tronlong@tronlong-virtual-machine:~/qt-demo/image_display/src$
```

图 5

2 基于 Qt Creator 工具编译 Qt 工程

请先自行下载、安装 Qt Creator 工具至 Ubuntu，执行如下命令启动 Qt Creator（本次操作使用 Qt Creator 版本为 4.11.0）。

Host# qtcreator

```
tronlong@tronlong-virtual-machine:~$ qtcreator
```

图 6

启动 Qt Creator 工具后，点击菜单栏"Tools -> Options..."。

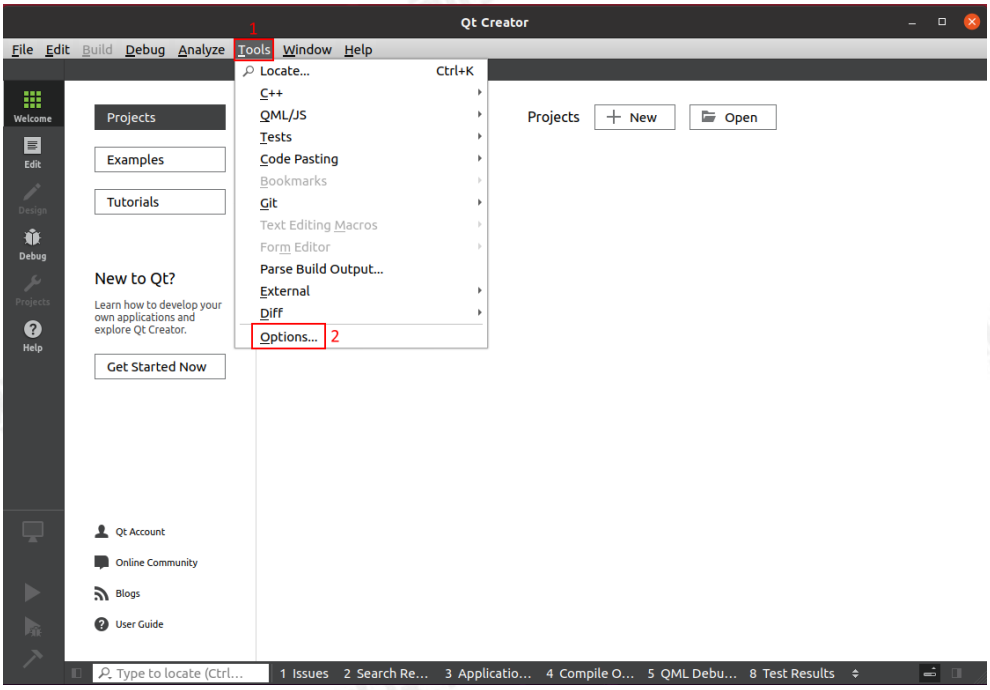

图 7

点击"Kits"，打开配置选择页面。

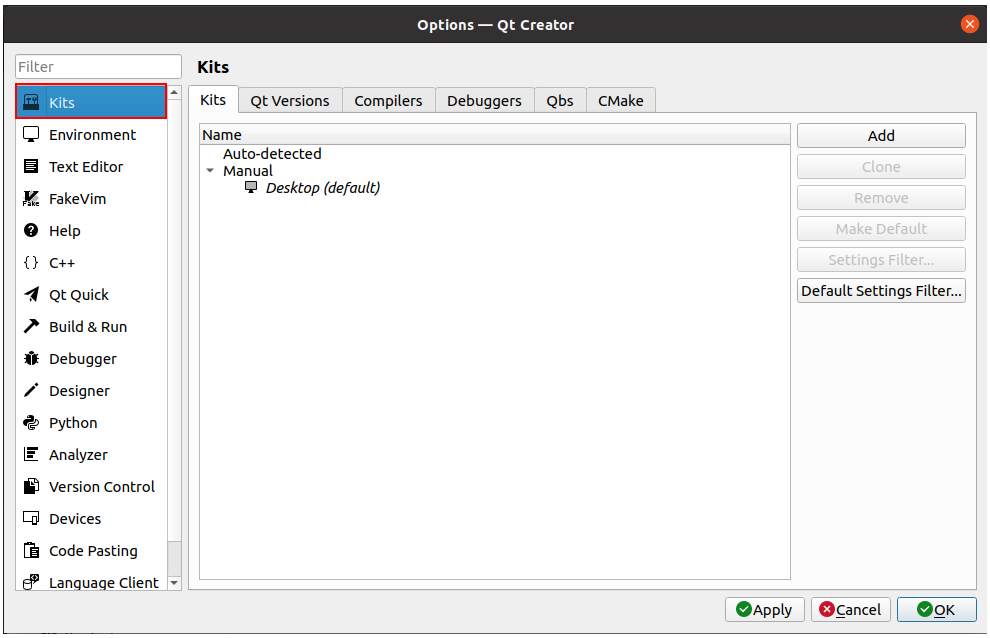

图 8

2.1 配置交叉编译工具链

点击"Compilers -> Add -> GCC -> C++ -> Browse...",选择平台交叉编译工具链中的 aarch64-poky-linux-g++，然后点击 Open 确认。具体路径为：

/home/tronlong/SDK/sysroots/x86_64-pokysdk-linux/usr/bin/aarch64-poky-linux/aarch64-poky-linux-g++

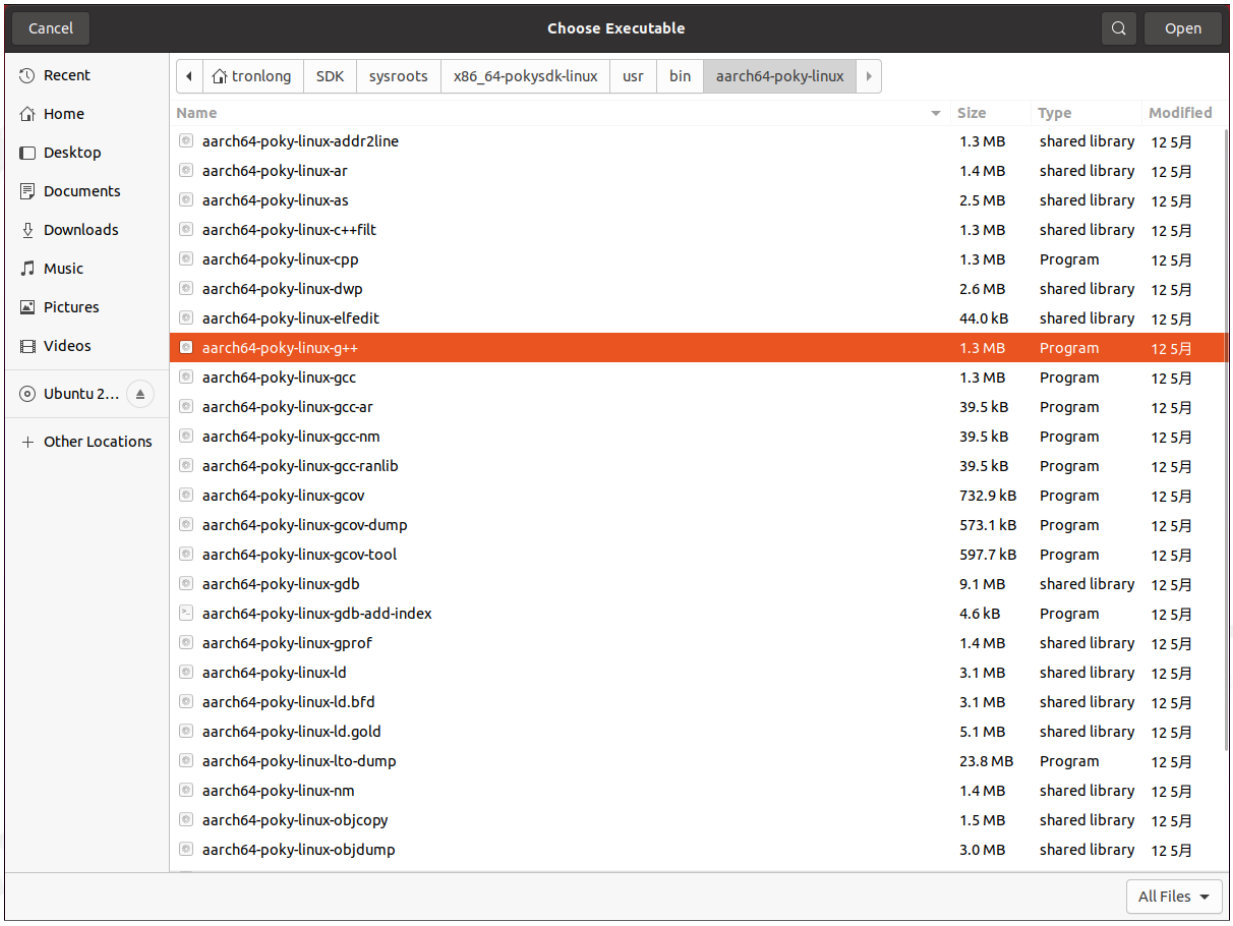

图 9

选择后，Name 修改为"IMX8MP-GCC"，ABI 选择"arm-linux-generic-elf-64bit"，再点击 Apply，如下图所示。

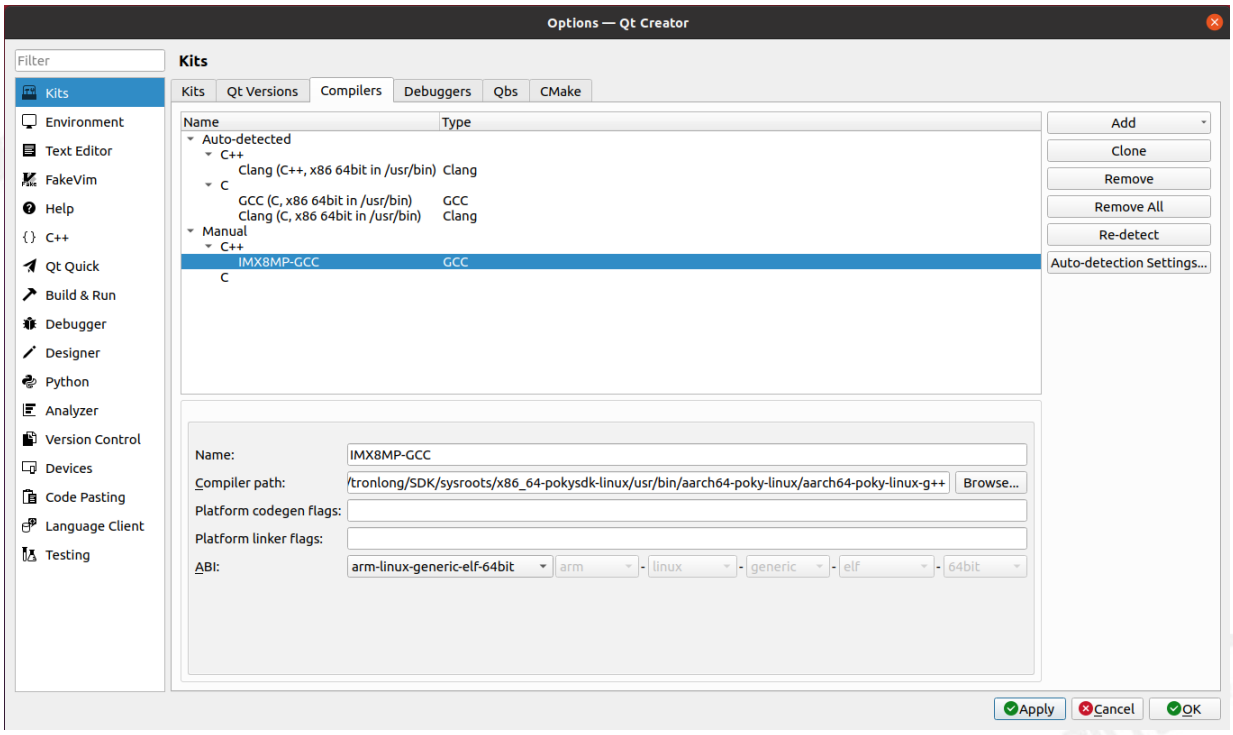

图 10

2.2 配置 Qt Versions

点击"Qt Versions -> Add"，选择平台 qmake，然后点击 Open 确认。具体路径为：/home/tronlong/SDK/sysroots/x86_64-pokysdk-linux/usr/bin/qmake

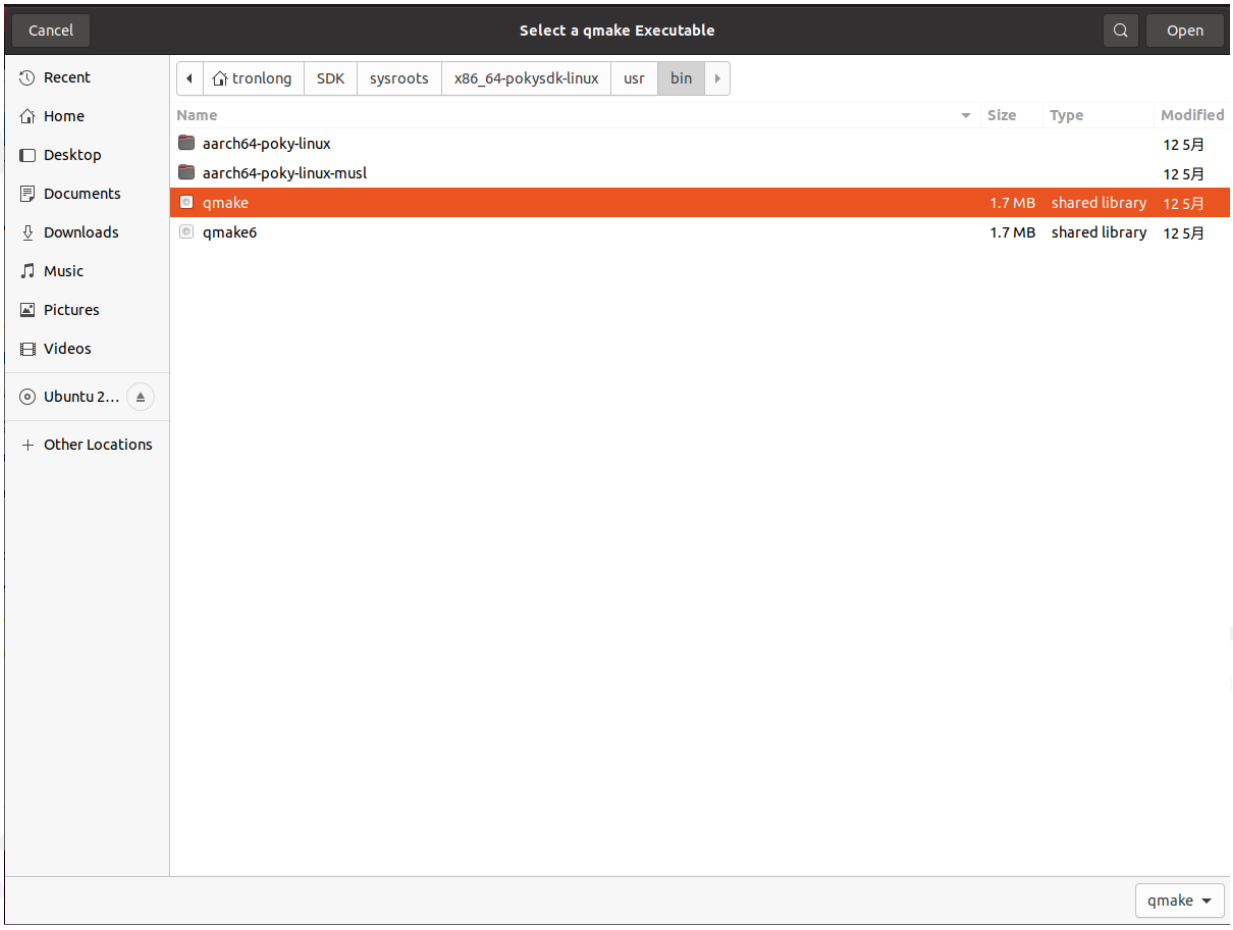

图 11

选择后，Version name 修改为"IMX8MP-qt6"再点击 Apply，如下图所示。

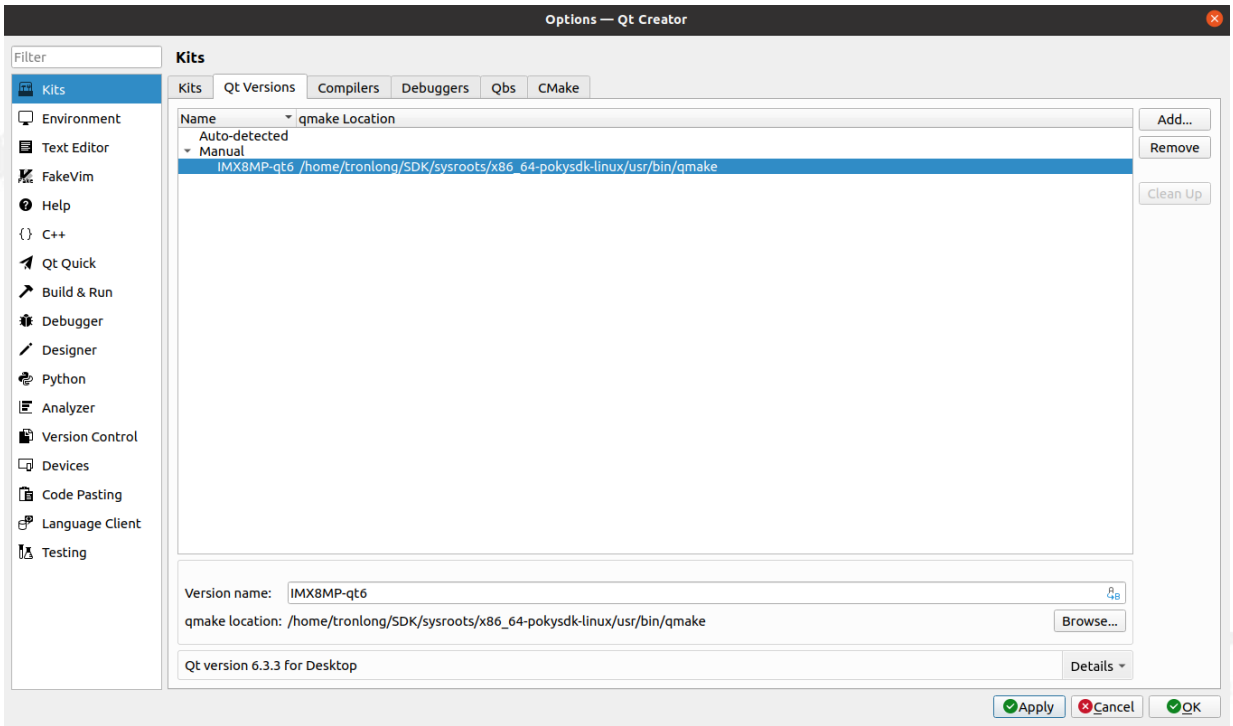

图 12

2.3 配置 Debuggers

点击"Debuggers -> Add -> Browse...", 选择平台 GDB，然后点击 Open 确认。具体路径为：

/home/tronlong/SDK/sysroots/x86_64-pokysdk-linux/usr/bin/aarch64-poky-linux/aarch64-poky-linux-gdb

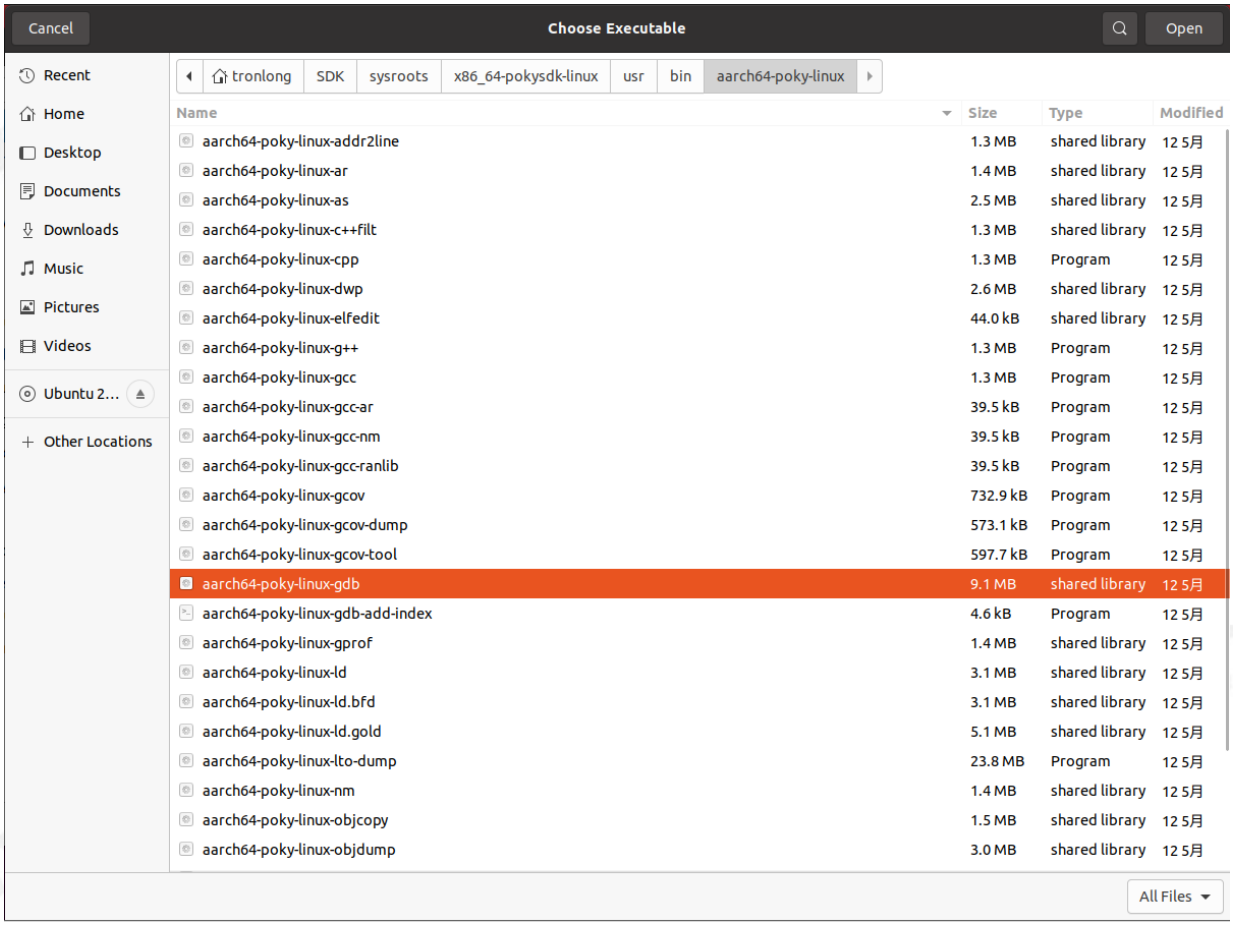

图 13

选择后，Name 修改为"IMX8MP-Debugger"再点击 Apply，如下图所示。

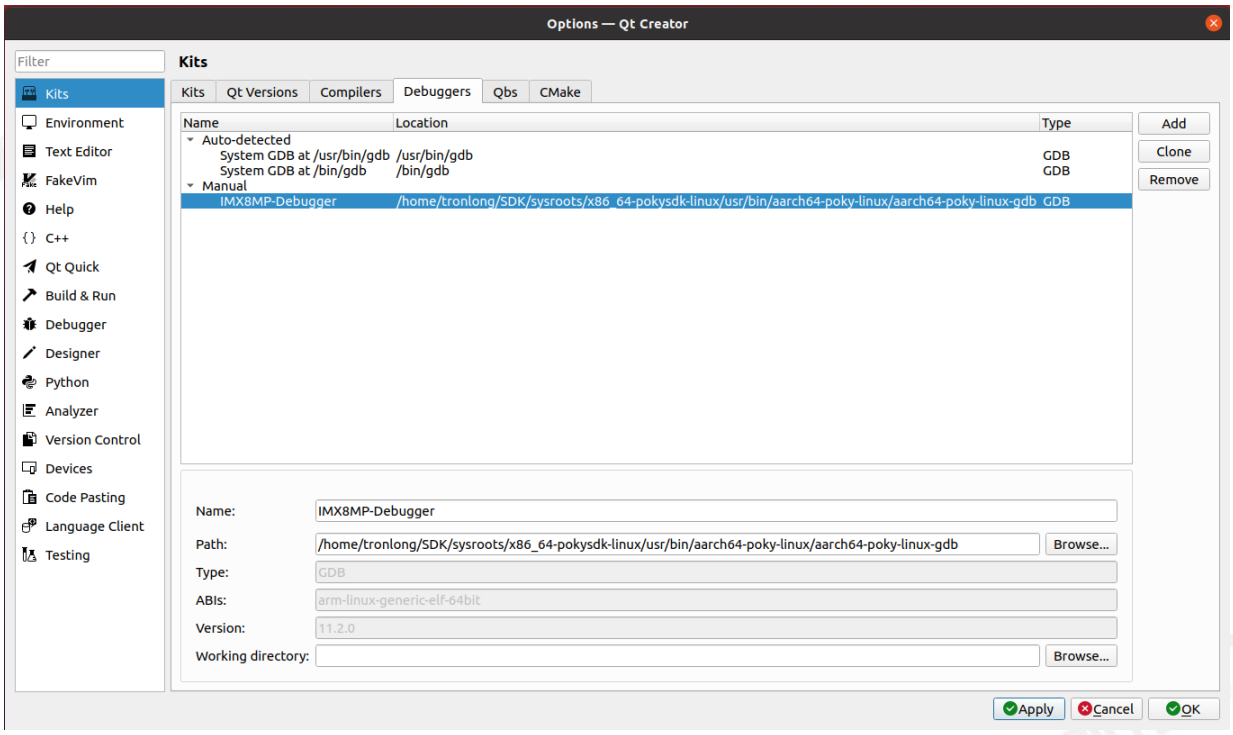

图 14

2.4 配置 Kits

请参照下图配置 Kits 各选项。点击"Kits -> Add"，将 Name 修改为"IMX8MP"，Device type 选择"Generic Linux Device"，Compiler 的 C++选择"IMX8MP-GCC"，Debugger 选择"IMX8MP-Debugger"，Qt version 选择"IMX8MP-qt6"，点击 Apply，然后点击 OK 完成配置。

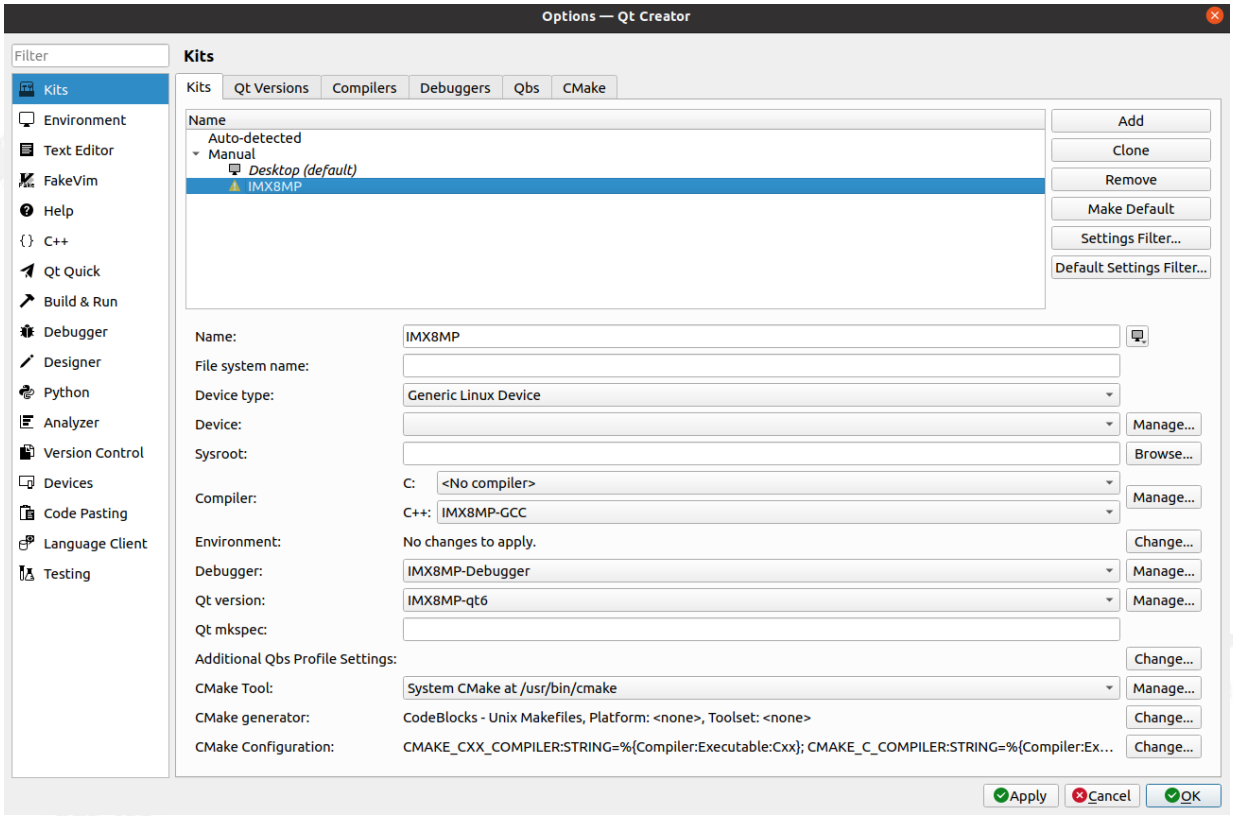

图 15

2.5 编译 Qt 工程

点击"Open"。

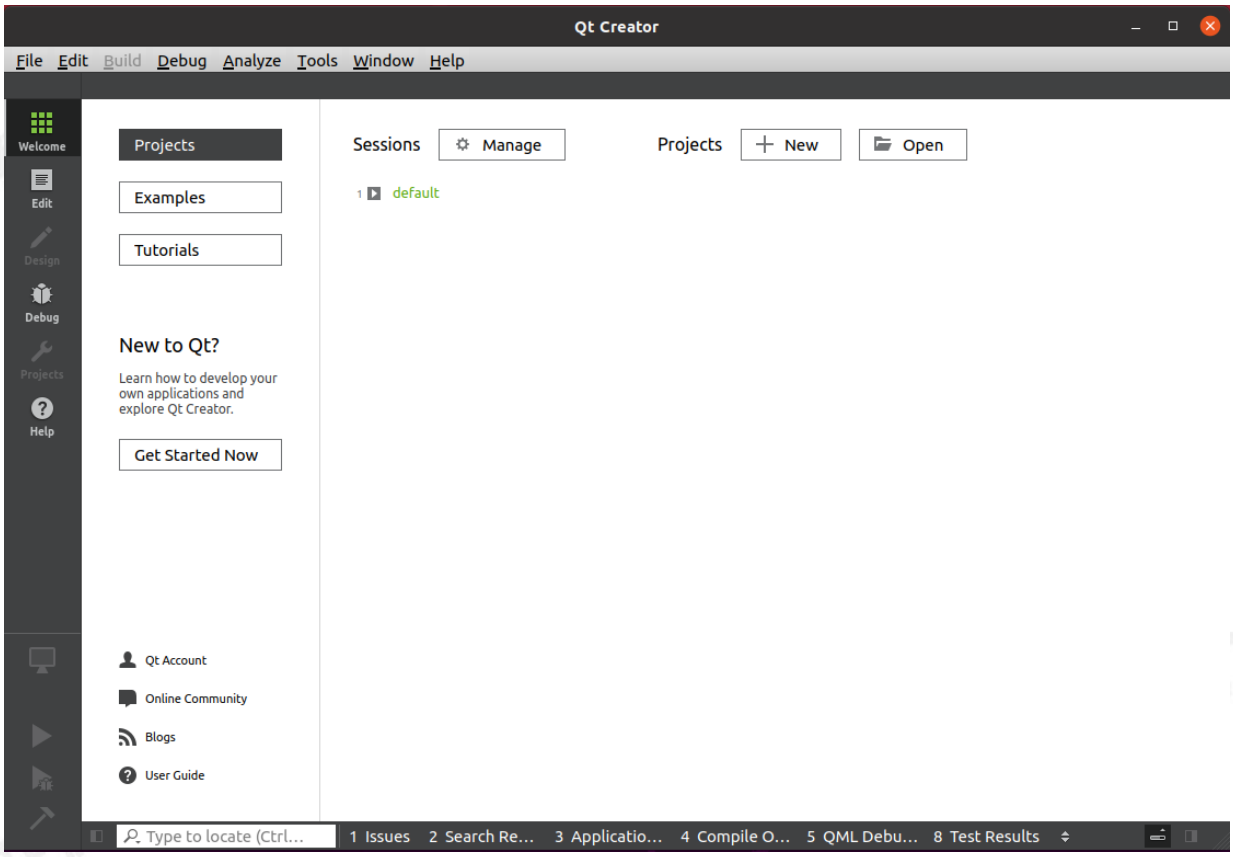

图 16

选中 Qt 工程源码目录中.pro 文件，点击 Open。

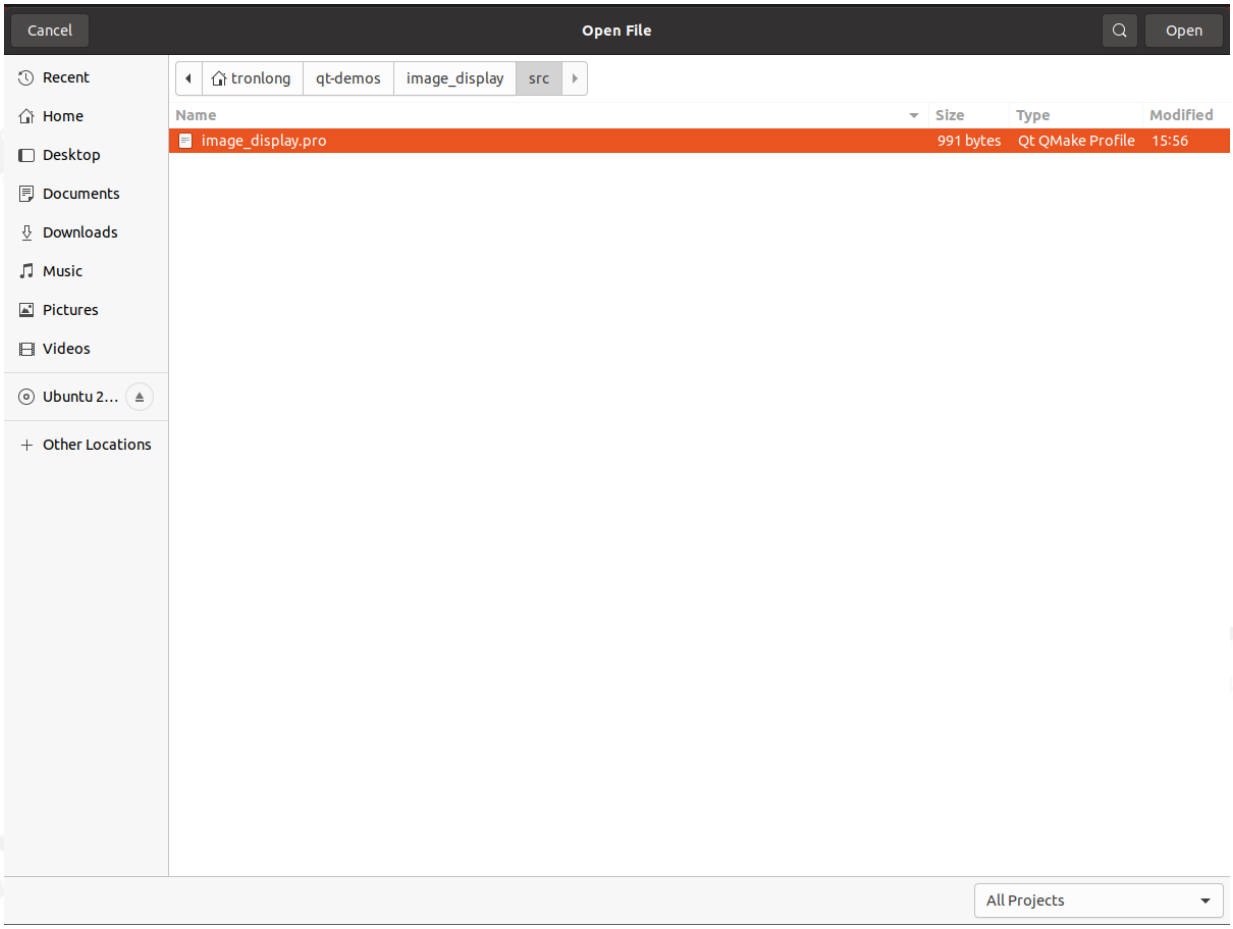

图 17

勾选 IMX8MP 平台 Kits，点击"Configure Project"进行工程配置，如下图所示。

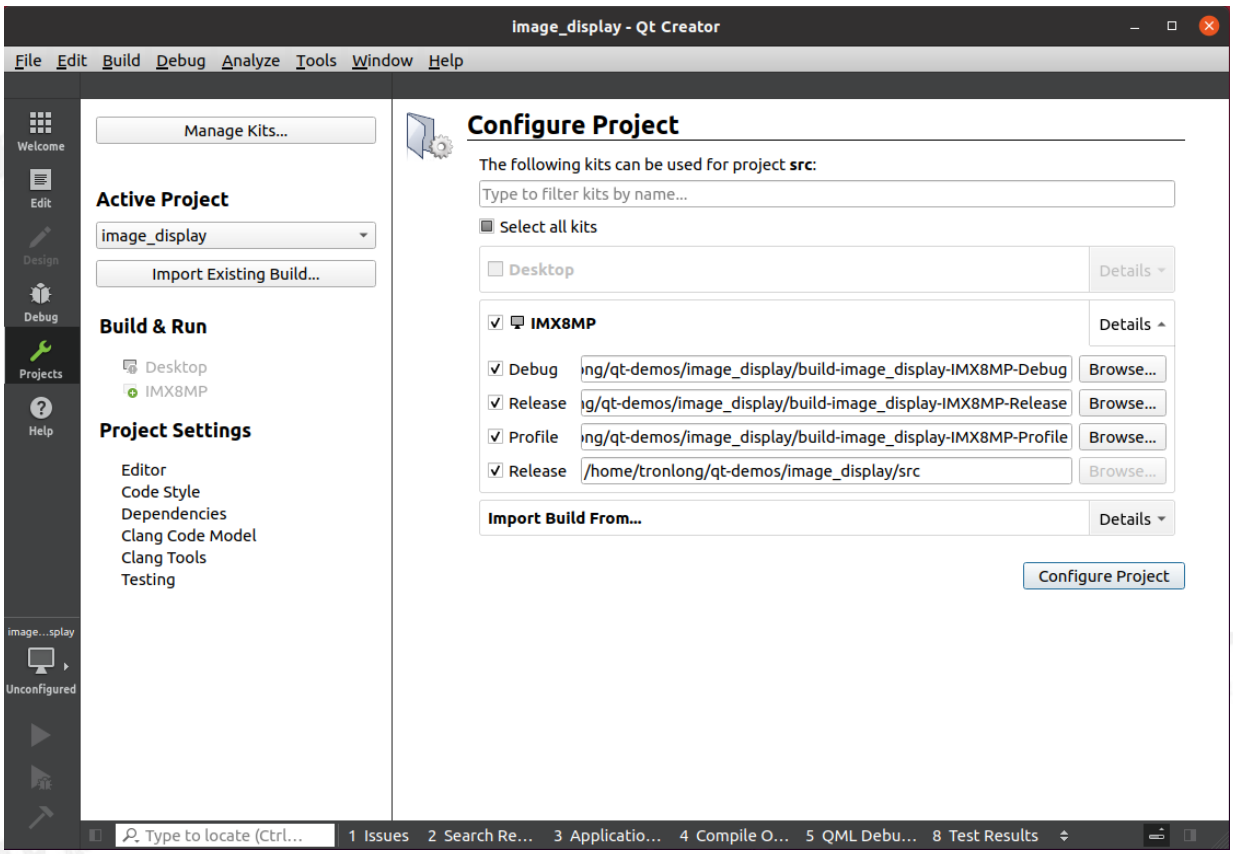

图 18

右击 Qt 工程，点击 Build 进行 Qt 工程编译。

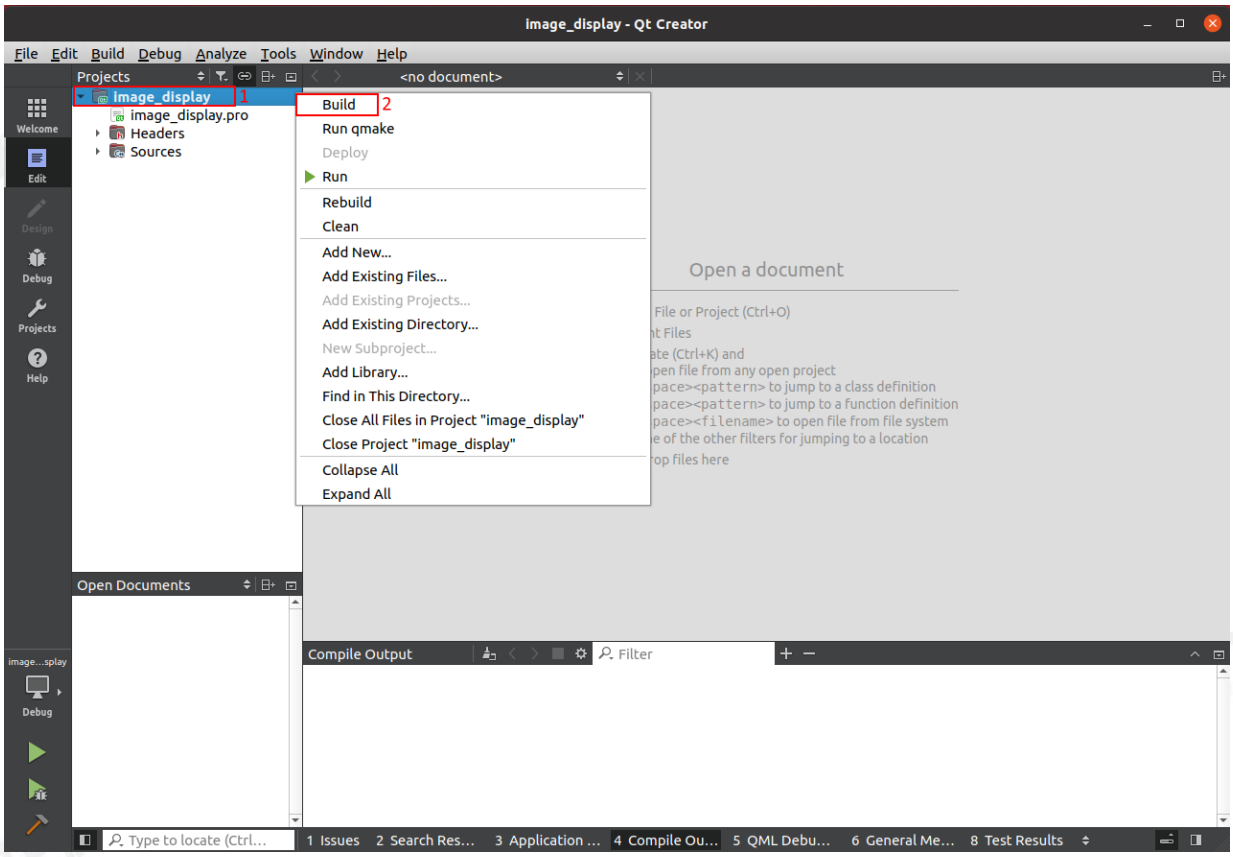
image_display - Qt Creator
File Edit Build Debug Analyze Tools Window Help
Projects
image_display
1
image_display.pro
Headers
Sources
Build
2
Run qmake
Deploy
Run
Rebuild
Clean
Add New...
Add Existing Files...
Add Existing Projects...
Add Existing Directory...
New Subproject...
Add Library...
Find in This Directory...
Close All Files in Project "image_display"
Close Project "image_display"
Collapse All
Expand All
Open a document
Open Documents
Compile Output
Welcome
Edit
Design
Debug
Projects
Help
1 Issues
2 Search Res...
3 Application ...
4 Compile Ou...
5 QML Debu...
6 General Me...
8 Test Results

图 19

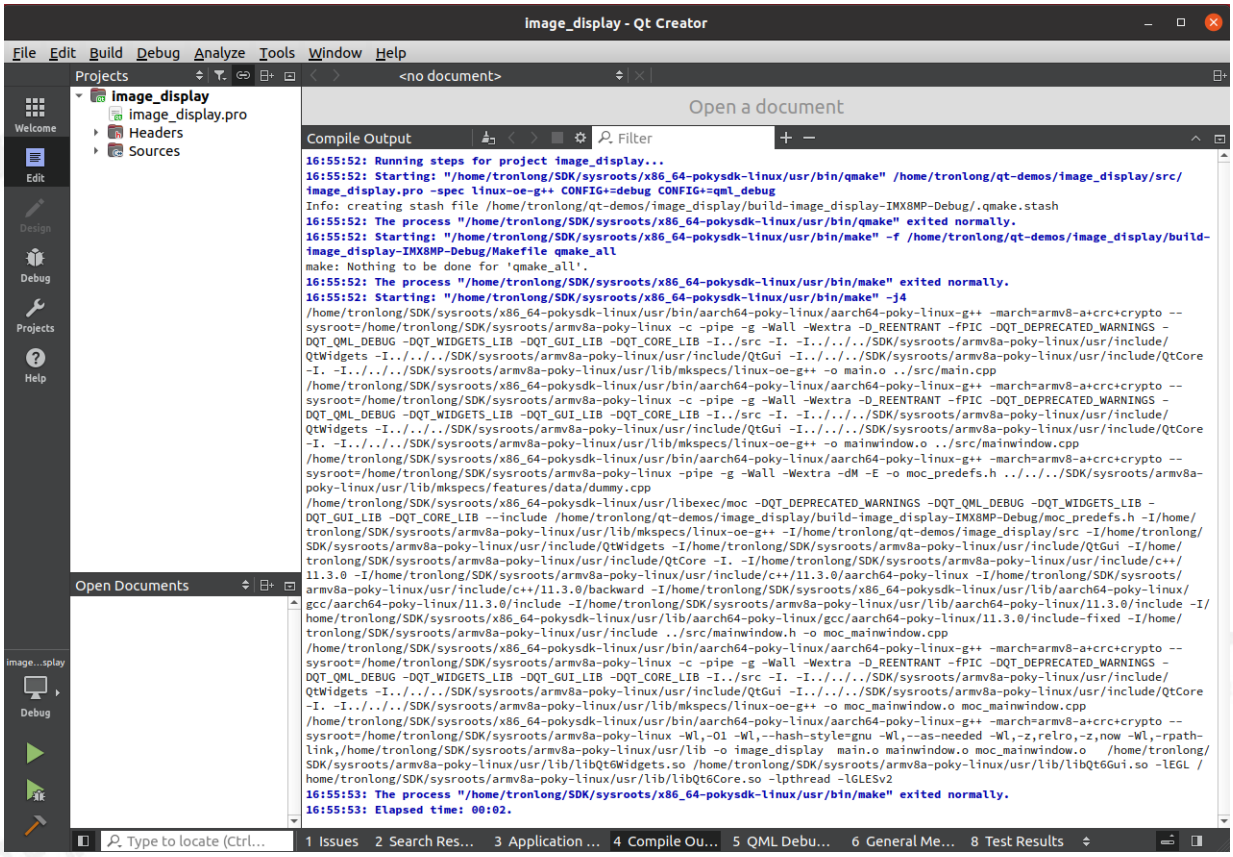

图 20

编译完成后，将生成可在评估板上正常运行的 ARM 端 Qt 程序镜像，如下图所示。

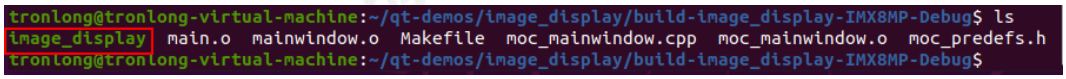

图 21

3 在评估板上运行 Qt 程序

评估板默认支持 HDMI1 OUT 和 HDMI2 OUT 接口显示，其中 HDMI1 OUT 为副显示，HDMI2 OUT 为主显示。本小节测试使用 HDMI2 OUT 接口，请通过 HDMI 线将 HDMI 显示屏与评估板 HDMI2 OUT 接口(CON22)连接。

如需使用 LVDS LCD 或 MIPI LCD 显示屏进行 Qt 程序测试，请参考《评估板测试手册》文档“LVDS LCD 接口测试”或“MIPI LCD 接口测试”章节将显示屏正确连接评估板，并切换相关设备树文件。

备注：如需使用 HDMI1 OUT 接口进行测试，请按如下步骤操作，若无需使用，请跳过此步骤。

评估板上电启动，进入评估板文件系统，执行如下命令，修改"/etc/xdg/weston/weston.ini"文件，将 Weston 配置中的 HDMI2 OUT 内部输出关闭。

Target# vi /etc/xdg/weston/weston.ini

```
root@IMX8MP-Tronlong:~#
root@IMX8MP-Tronlong:~# vi /etc/xdg/weston/weston.ini
```

图 22

添加内容如下：

[output]

name=DSI-1

mode=off

```
[core]
#gbm-format=argb8888
idle-time=0
use-g2d=1
#xwayland=true
repaint-window=16
#enable-overlay-view=1
modules=screen-share.so

#[shell]
#size=1920x1080

[libinput]
touchscreen_calibrator=true

[output]
name=DSI-1
mode=off

#[output]
#name=HDMI-A-1
#mode=1920x1080@60
#transform=rotate-90

#[output]
#name=HDMI-A-2
#mode=off
#       WIDTHxHEIGHT    Resolution size width and height in pixels
#       off             Disables the output
#       preferred       Uses the preferred mode
#       current         Uses the current crt controller mode
#transform=rotate-90

[screen-share]
command=/usr/bin/weston --backend=rdp-backend.so --shell=fullscreen-shell.so --no-clients-resize --r
#start-on-startup=true

[launcher]
icon=/home/root/.nxp-demo-experience/icon/icon_demo_launcher.png
path=/usr/bin/demoexperience

[launcher]
icon=/usr/share/weston/terminal.png
path=/usr/bin/weston-terminal
```

图 23

保存后退出，执行如下命令重新启动 Weston。

Target# systemctl restart weston

```
root@IMX8MP-Tronlong:~# systemctl restart weston
[   63.492530] audit: type=1006 audit(1651170787.212:5): pid=942 uid=0 old-auid=4294967295
 auid=0 tty=tty7 old-ses=4294967295 ses=4 res=1
[   63.504703] audit: type=1300 audit(1651170787.212:5): arch=c00000b7 syscall=64 success=
yes exit=1 a0=8 a1=ffffc8bc4310 a2=1 a3=0 items=0 ppid=1 pid=942 auid=0 uid=0 gid=0 euid=0
 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=tty7 ses=4 comm="(weston)" exe="/lib/systemd/sys
temd" key=(null)
[   63.529852] audit: type=1327 audit(1651170787.212:5): proctitle="(weston)"
root@IMX8MP-Tronlong:~#
```

图 24

将 ARM 端 Qt 程序镜像拷贝至评估板文件系统任意目录下，同时将 image_display 工程 bin 目录下文件名为 test.jpg 的 JPG 格式图片拷贝至相同路径。请在 Qt 程序镜像所在路径下执行如下命令。

（1） 查看程序运行参数。

Target# ./image_display

参数解释

image path：图片路径。

rotation：该参数为 1，则竖屏显示。

```
root@IMX8MP-Tronlong:~# ./image_display
usage : ./image_display <image path> [rotation]
        e.g. : ./image_display ./test.jpg
        e.g. : ./image_display ./test.jpg 1
root@IMX8MP-Tronlong:~#
```

图 25

（2） 横屏显示图片。

执行如下命令，显示屏将显示如下图像，按"Ctrl + C"退出程序。

Target# ./image_display ./test.jpg

```
root@IMX8MP-Tronlong:~# ./image_display ./test.jpg
```

图 26

图 27

（3） 竖屏显示图片。

执行如下命令，显示屏将显示如下图像，按"Ctrl + C"退出程序。

Target# ./image_display test.jpg 1

```
root@IMX8MP-Tronlong:~# ./image_display test.jpg 1
```

图 28

图 29

4 Qt 程序自启动说明

本章节以 led_control 案例为例，演示 Qt 程序自启动功能。

评估板默认支持 HDMI1 OUT 和 HDMI2 OUT 接口显示，其中 HDMI1 OUT 为副显示，HDMI2 OUT 为主显示。请将 HDMI 显示器与评估板 HDMI2 OUT 接口(CON22)连接，同时将鼠标与评估板 USB3.0 HOST1(USB2)、USB3.0 HOST2(USB2)或 USB3.0 HOST3(USB2)任一接口连接。

进入评估板文件系统，执行如下命令，新建 Qt 程序自启动指定配置文件。

Target# vi /etc/default/qtenv

```
root@IMX8MP-Tronlong:~# vi /etc/default/qtenv
root@IMX8MP-Tronlong:~# cat /etc/default/qtenv
XDG_RUNTIME_DIR=/run/user/0
QT_QPA_PLATFORM=wayland
root@IMX8MP-Tronlong:~#
```

图 30

配置文件内容如下：

XDG_RUNTIME_DIR=/run/user/0

QT_QPA_PLATFORM=wayland

```
XDG_RUNTIME_DIR=/run/user/0
QT_QPA_PLATFORM=wayland
```

图 31

执行如下命令，在"/etc/systemd/system/"目录下新建 Qt 程序自启动服务依赖环境和配置程序路径的文件。

Target# vi /etc/systemd/system/qtapp.service

```
root@IMX8MP-Tronlong:~# vi /etc/systemd/system/qtapp.service
root@IMX8MP-Tronlong:~# cat /etc/systemd/system/qtapp.service
[Unit]
Description=Qt App Service
Requisite=weston.service
After=weston.service

[Service]
EnvironmentFile=/etc/default/qtenv
ExecStart=/home/root/led_control

[Install]
WantedBy=graphical.target
root@IMX8MP-Tronlong:~#
```

图 32

```
[Unit]
Description=Qt App Service
Requisite=weston.service
After=weston.service

[Service]
EnvironmentFile=/etc/default/qtenv
ExecStart=/home/root/led_control

[Install]
WantedBy=graphical.target
```

图 33

文件内容如下：

[Unit]

Description=Qt App Service //服务程序描述为 Qt 应用程序

Requisite=weston.service //当前服务程序依赖与 weston 服务

After=weston.service //weston 服务启动完成后，当前服务才启动

[Service]

EnvironmentFile=/etc/default/qtenv　　//指定 Qt 程序启动环境变量配置文件

ExecStart=/home/root/led_control　　//指定服务启动 Qt 程序路径

[Install]

WantedBy=graphical.target　　//添加到 graphical.target 单元中进行管理

执行如下命令，使能 qtapp 程序自启动服务，评估板重启后，led_control 程序将会自启动。

Target#　systemctl enable qtapp

Target#　reboot

备注：警告信息可能与 systemd 的特性有关，不影响功能使用。

```
root@IMX8MP-Tronlong:~# systemctl enable qtapp
Created symlink /etc/systemd/system/graphical.target.wants/qtapp.service → /etc/systemd/syst
em/qtapp.service.
[  714.422488] systemd-sysv-generator[705]: SysV service '/etc/init.d/sendsigs' lacks a nati
ve systemd unit file. Automatically generating a unit file for compatibility. Please update
package to include a native systemd unit file, in order to make it more safe and robust.
[  714.422610] systemd-sysv-generator[705]: SysV service '/etc/init.d/single' lacks a native
 systemd unit file. Automatically generating a unit file for compatibility. Please update pa
ckage to include a native systemd unit file, in order to make it more safe and robust.
[  714.423099] systemd-sysv-generator[705]: SysV service '/etc/init.d/reboot' lacks a native
 systemd unit file. Automatically generating a unit file for compatibility. Please update pa
ckage to include a native systemd unit file, in order to make it more safe and robust.
[  714.423542] systemd-sysv-generator[705]: SysV service '/etc/init.d/sysrepo-tsnd' lacks a
native systemd unit file. Automatically generating a unit file for compatibility. Please upd
ate package to include a native systemd unit file, in order to make it more safe and robust.
[  714.423650] systemd-sysv-generator[705]: SysV service '/etc/init.d/save-rtc.sh' lacks a n
ative systemd unit file. Automatically generating a unit file for compatibility. Please upda
te package to include a native systemd unit file, in order to make it more safe and robust.
[  714.423760] systemd-sysv-generator[705]: SysV service '/etc/init.d/sysrepod' lacks a nati
ve systemd unit file. Automatically generating a unit file for compatibility. Please update
package to include a native systemd unit file, in order to make it more safe and robust.
[  714.423863] systemd-sysv-generator[705]: SysV service '/etc/init.d/umountfs' lacks a nati
ve systemd unit file. Automatically generating a unit file for compatibility. Please update
package to include a native systemd unit file, in order to make it more safe and robust.
[  714.424006] systemd-sysv-generator[705]: SysV service '/etc/init.d/umountnfs.sh' lacks a
native systemd unit file. Automatically generating a unit file for compatibility. Please upd
ate package to include a native systemd unit file, in order to make it more safe and robust.
[  714.424107] systemd-sysv-generator[705]: SysV service '/etc/init.d/rc.local' lacks a nati
ve systemd unit file. Automatically generating a unit file for compatibility. Please update
package to include a native systemd unit file, in order to make it more safe and robust.
[  714.424407] systemd-sysv-generator[705]: SysV service '/etc/init.d/sysrepo-init' lacks a
native systemd unit file. Automatically generating a unit file for compatibility. Please upd
ate package to include a native systemd unit file, in order to make it more safe and robust.
[  714.424684] systemd-sysv-generator[705]: SysV service '/etc/init.d/halt' lacks a native s
ystemd unit file. Automatically generating a unit file for compatibility. Please update pack
age to include a native systemd unit file, in order to make it more safe and robust.
[  714.424888] systemd-sysv-generator[705]: SysV service '/etc/init.d/netopeer2-server' lack
s a native systemd unit file. Automatically generating a unit file for compatibility. Please
 update package to include a native systemd unit file, in order to make it more safe and rob
ust.
[  714.425528] systemd-sysv-generator[705]: SysV service '/etc/init.d/sysrepo-plugind' lacks
 a native systemd unit file. Automatically generating a unit file for compatibility. Please
update package to include a native systemd unit file, in order to make it more safe and robu
st.
root@IMX8MP-Tronlong:~# reboot
```

图 34

Qt 程序运行成功后，可观察到 HDMI 显示屏显示 LED 控制界面，可通过鼠标点击界面控制 LED 亮灭，如下图所示。如需通过触摸方式控制 LED，请切换为 MIPI LCD、LVDS LCD 等显示方式。

图 35

如需取消评估板 Qt 程序自启动，请执行如下命令关闭 qtapp 程序自启动服务。

Target# systemctl disable qtapp

Target# reboot

备注：警告信息可能与 systemd 的特性有关，不影响功能使用。

```
root@IMX8MP-Tronlong:~# systemctl disable qtapp
Removed /etc/systemd/system/graphical.target.wants/qtapp.service.
[  123.391085] systemd-sysv-generator[666]: SysV service '/etc/init.d/sendsigs' lacks a nati
ve systemd unit file. Automatically generating a unit file for compatibility. Please update
package to include a native systemd unit file, in order to make it more safe and robust.
[  123.391204] systemd-sysv-generator[666]: SysV service '/etc/init.d/single' lacks a native
 systemd unit file. Automatically generating a unit file for compatibility. Please update pa
ckage to include a native systemd unit file, in order to make it more safe and robust.
[  123.391641] systemd-sysv-generator[666]: SysV service '/etc/init.d/reboot' lacks a native
 systemd unit file. Automatically generating a unit file for compatibility. Please update pa
ckage to include a native systemd unit file, in order to make it more safe and robust.
[  123.392100] systemd-sysv-generator[666]: SysV service '/etc/init.d/sysrepo-tsnd' lacks a
native systemd unit file. Automatically generating a unit file for compatibility. Please upd
ate package to include a native systemd unit file, in order to make it more safe and robust.
[  123.392246] systemd-sysv-generator[666]: SysV service '/etc/init.d/save-rtc.sh' lacks a n
ative systemd unit file. Automatically generating a unit file for compatibility. Please upda
te package to include a native systemd unit file, in order to make it more safe and robust.
[  123.392366] systemd-sysv-generator[666]: SysV service '/etc/init.d/sysrepod' lacks a nati
ve systemd unit file. Automatically generating a unit file for compatibility. Please update
package to include a native systemd unit file, in order to make it more safe and robust.
[  123.392481] systemd-sysv-generator[666]: SysV service '/etc/init.d/umountfs' lacks a nati
ve systemd unit file. Automatically generating a unit file for compatibility. Please update
package to include a native systemd unit file, in order to make it more safe and robust.
[  123.392622] systemd-sysv-generator[666]: SysV service '/etc/init.d/umountnfs.sh' lacks a
native systemd unit file. Automatically generating a unit file for compatibility. Please upd
ate package to include a native systemd unit file, in order to make it more safe and robust.
[  123.395404] systemd-sysv-generator[666]: SysV service '/etc/init.d/rc.local' lacks a nati
ve systemd unit file. Automatically generating a unit file for compatibility. Please update
package to include a native systemd unit file, in order to make it more safe and robust.
[  123.395714] systemd-sysv-generator[666]: SysV service '/etc/init.d/sysrepo-init' lacks a
native systemd unit file. Automatically generating a unit file for compatibility. Please upd
ate package to include a native systemd unit file, in order to make it more safe and robust.
[  123.395957] systemd-sysv-generator[666]: SysV service '/etc/init.d/halt' lacks a native s
ystemd unit file. Automatically generating a unit file for compatibility. Please update pack
age to include a native systemd unit file, in order to make it more safe and robust.
[  123.396069] systemd-sysv-generator[666]: SysV service '/etc/init.d/netopeer2-server' lack
s a native systemd unit file. Automatically generating a unit file for compatibility. Please
 update package to include a native systemd unit file, in order to make it more safe and rob
ust.
[  123.396554] systemd-sysv-generator[666]: SysV service '/etc/init.d/sysrepo-plugind' lacks
 a native systemd unit file. Automatically generating a unit file for compatibility. Please
update package to include a native systemd unit file, in order to make it more safe and robu
st.
root@IMX8MP-Tronlong:~# reboot
```

图 36

更多帮助

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